

Chemokine Receptor Taxonomical Analysis

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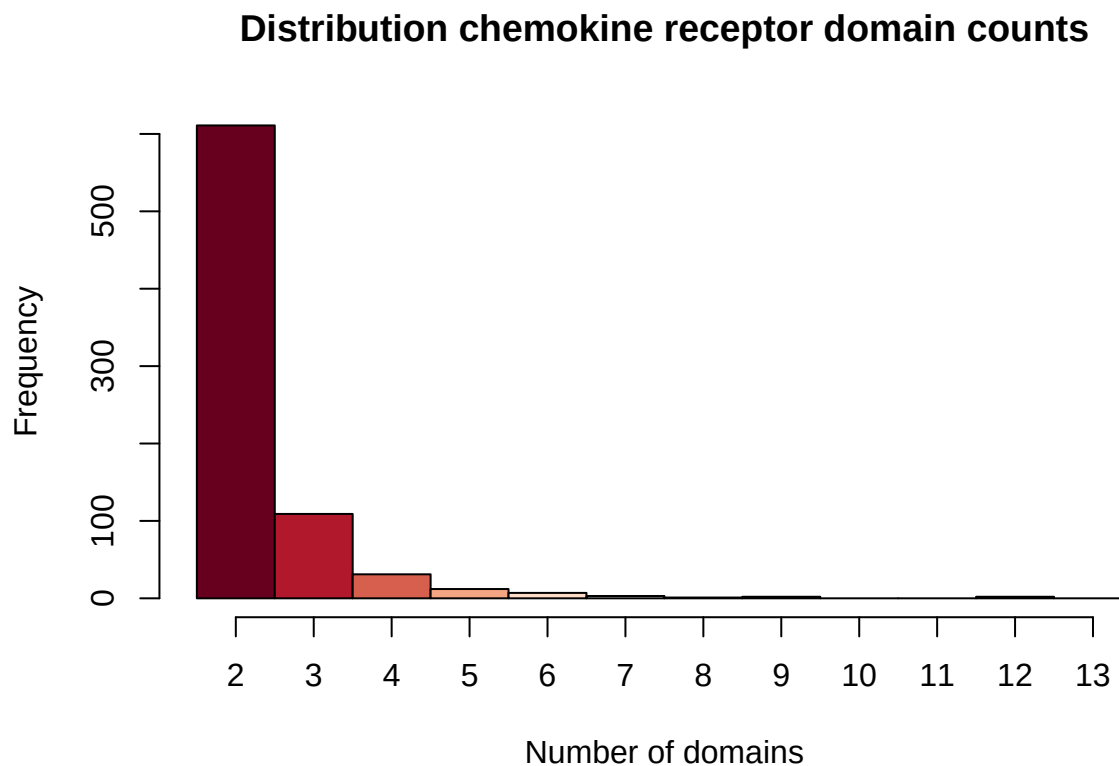
Taxonomical Analysis

First look

First we get the TSV file from the HMMSEARCH output parser. This file contains names of proteins (e.g. UniRef90_A0A1B0CRB2), amount of chemokine receptor domain counts, and lastly the taxonomic identifier (e.g. 9606 is for “Homo Sapiens”)

Domains

Here, we can see that we have 778 unique proteins with a minimum of 2 and a maximum of 12 domain hits. The distribution of the number of domain frequencies are as follows:



Since the domain count distribution is heavily skewed to the left (78.5347044% of proteins have 2 domains), I will be ignoring 2+ domains for the first part. We can address them later.

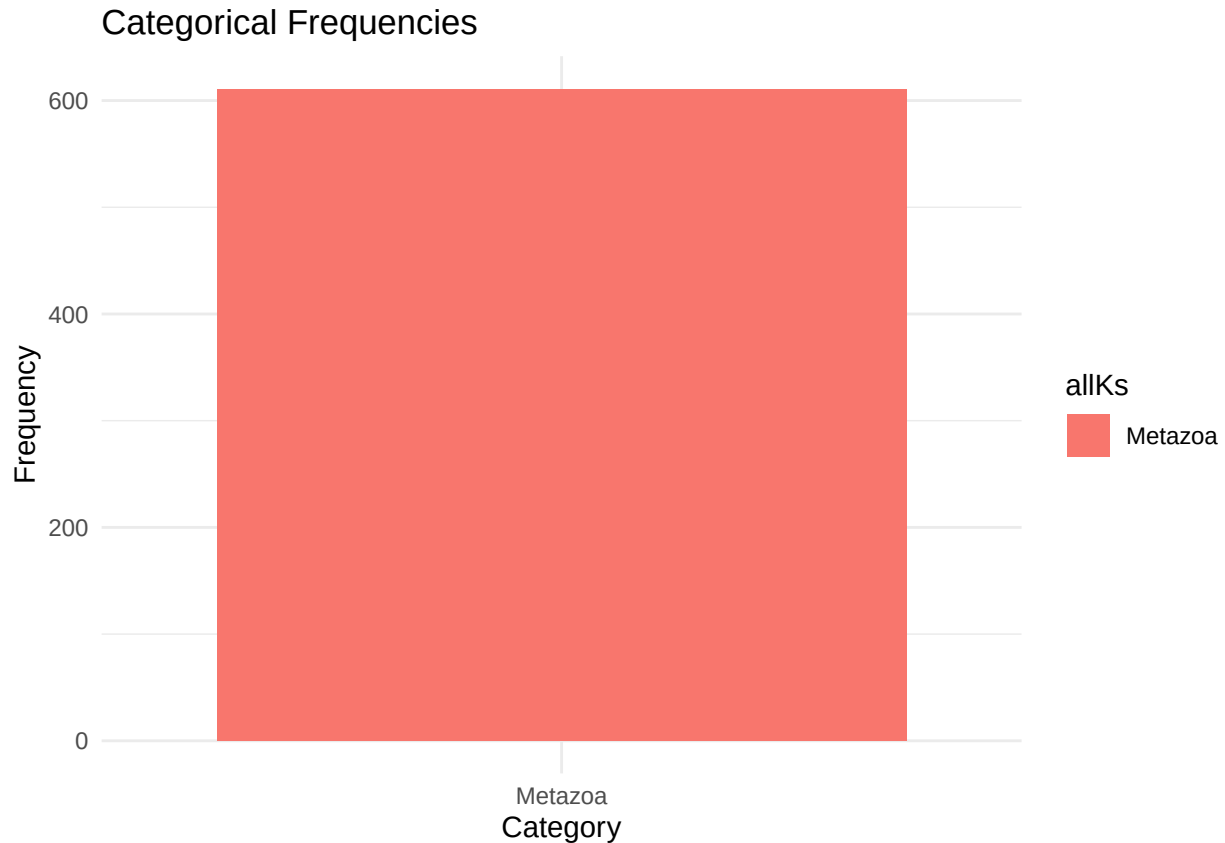
High-level taxonomy (Superkingdoms, kingdoms, and phylum)

Superkingdoms

- There are 1 superkingdom(s), namely: . Eukaryotes represent 100 (There are 0 from Bacteria).

Kingdoms

- There are 1 kingdoms, namely: . Here is the distribution:

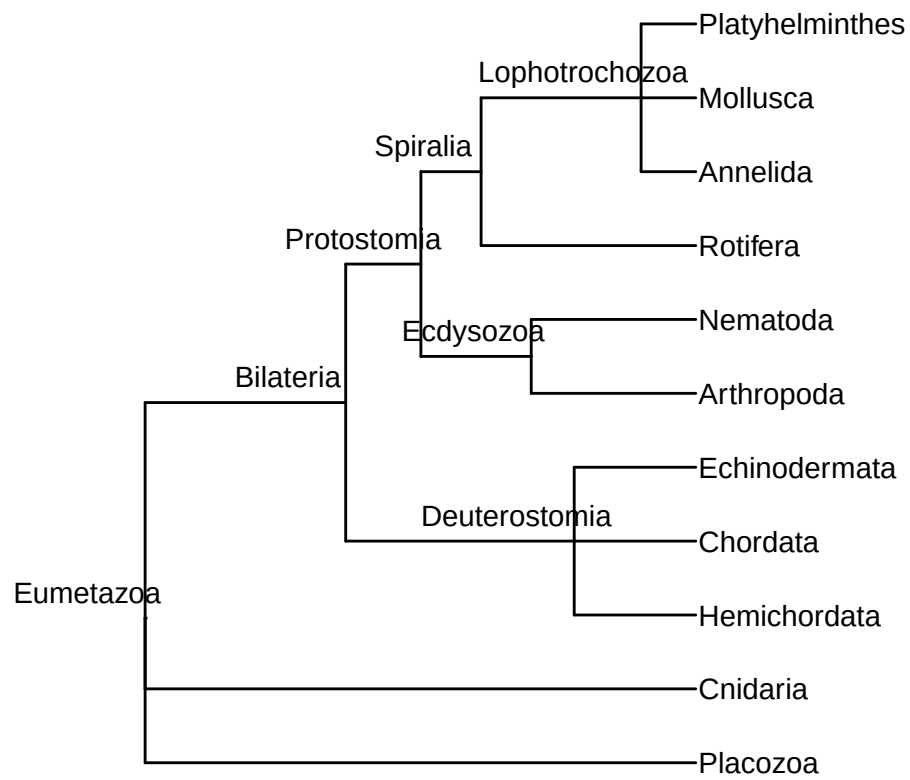


Phylum

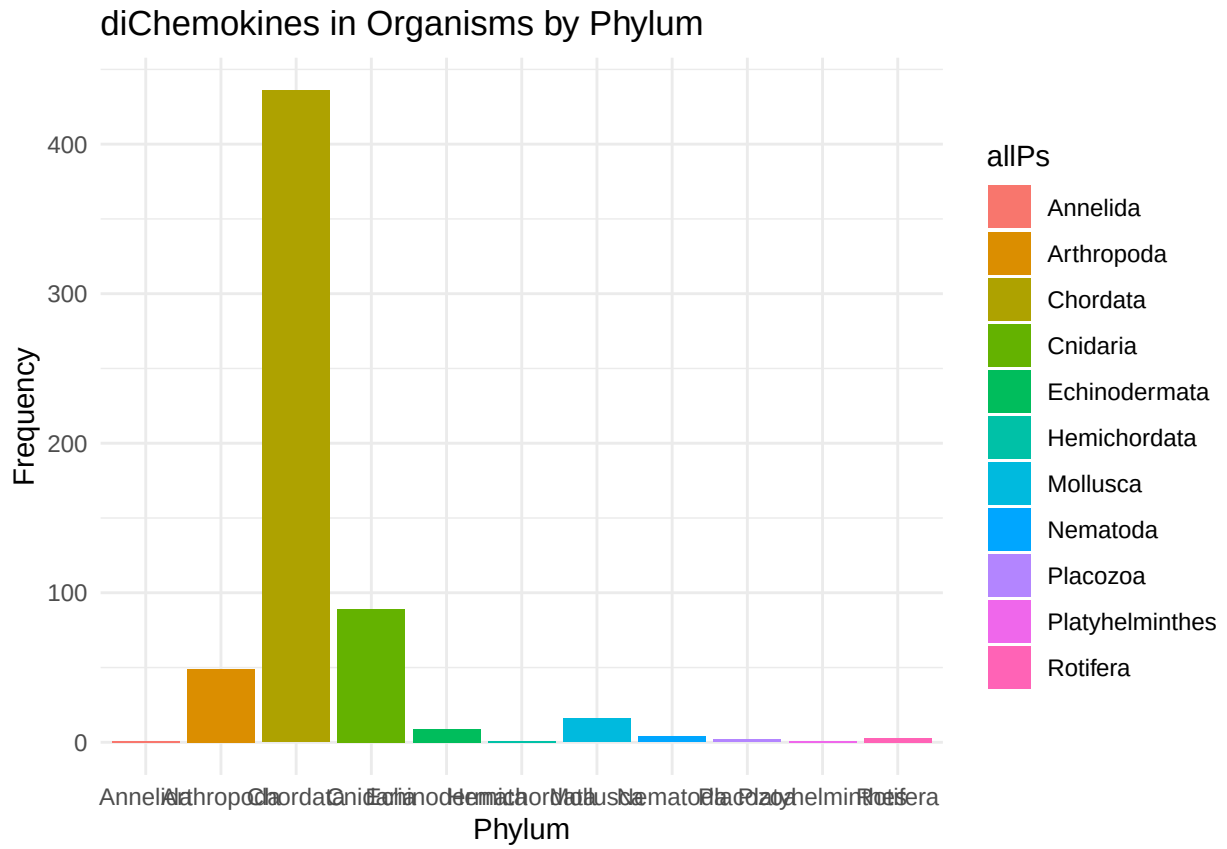
- There are 11 phylums, namely: .

Here is a visual representation of all the phylums present (which covers all the lower-level taxa):

```
## Get all ranks and their taxIDs
## Align taxonomy hierarchies...
## Taxonomy alignment done!
## Calculate distance matrix
## Add node labels
```

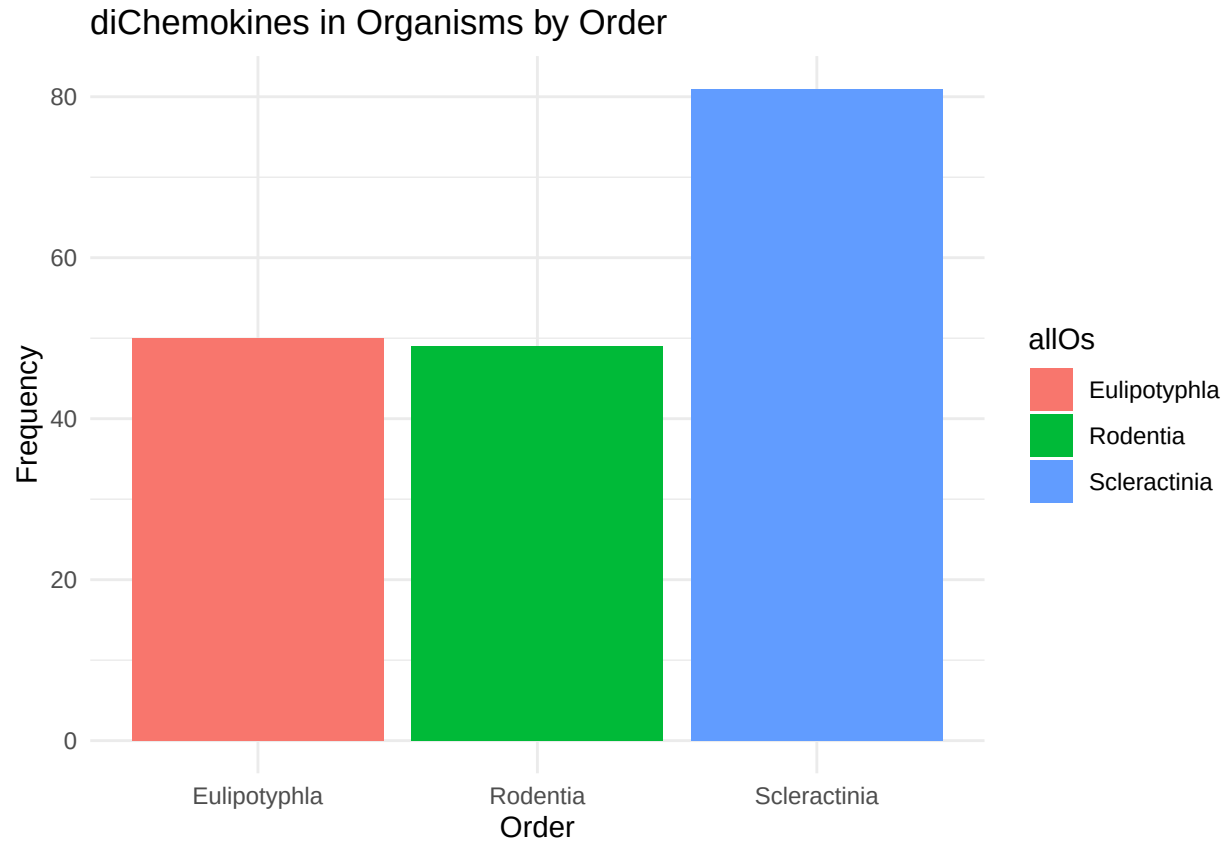


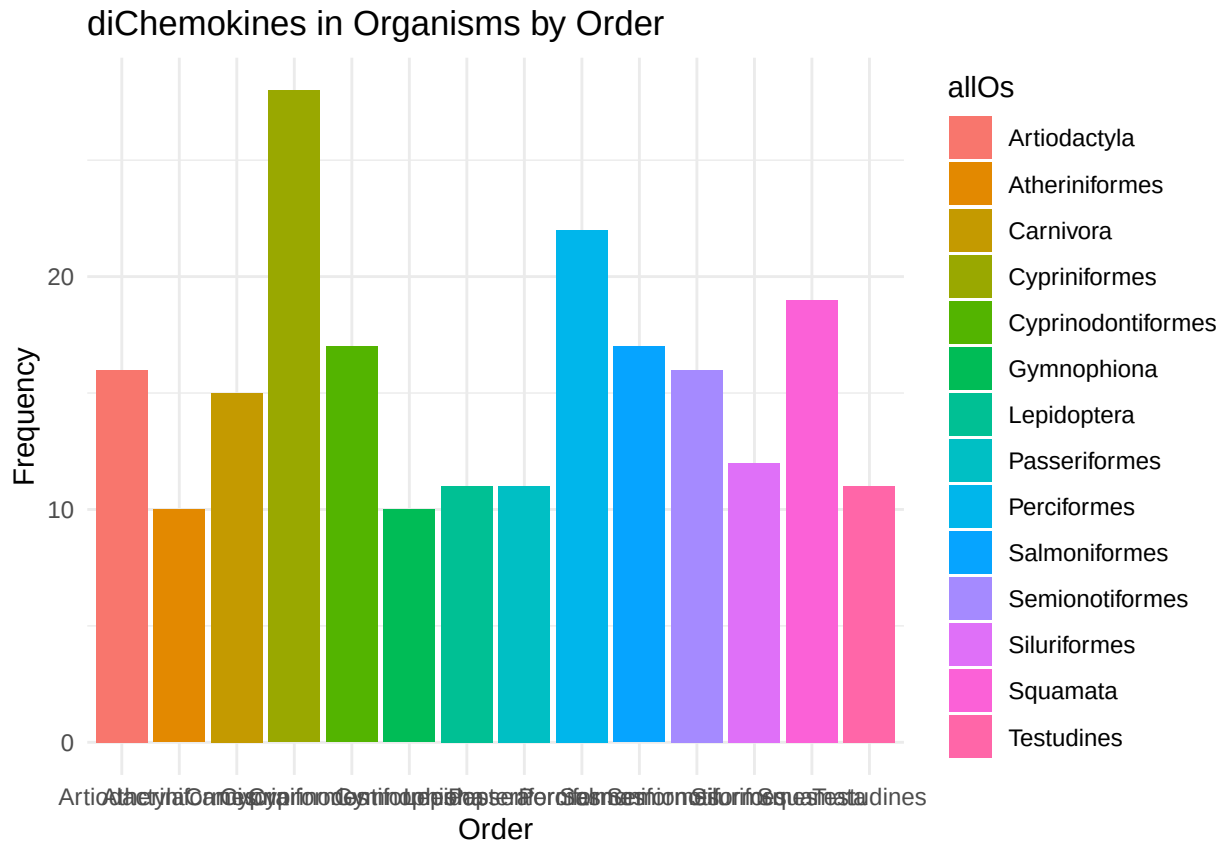
Also the histogram distribution:

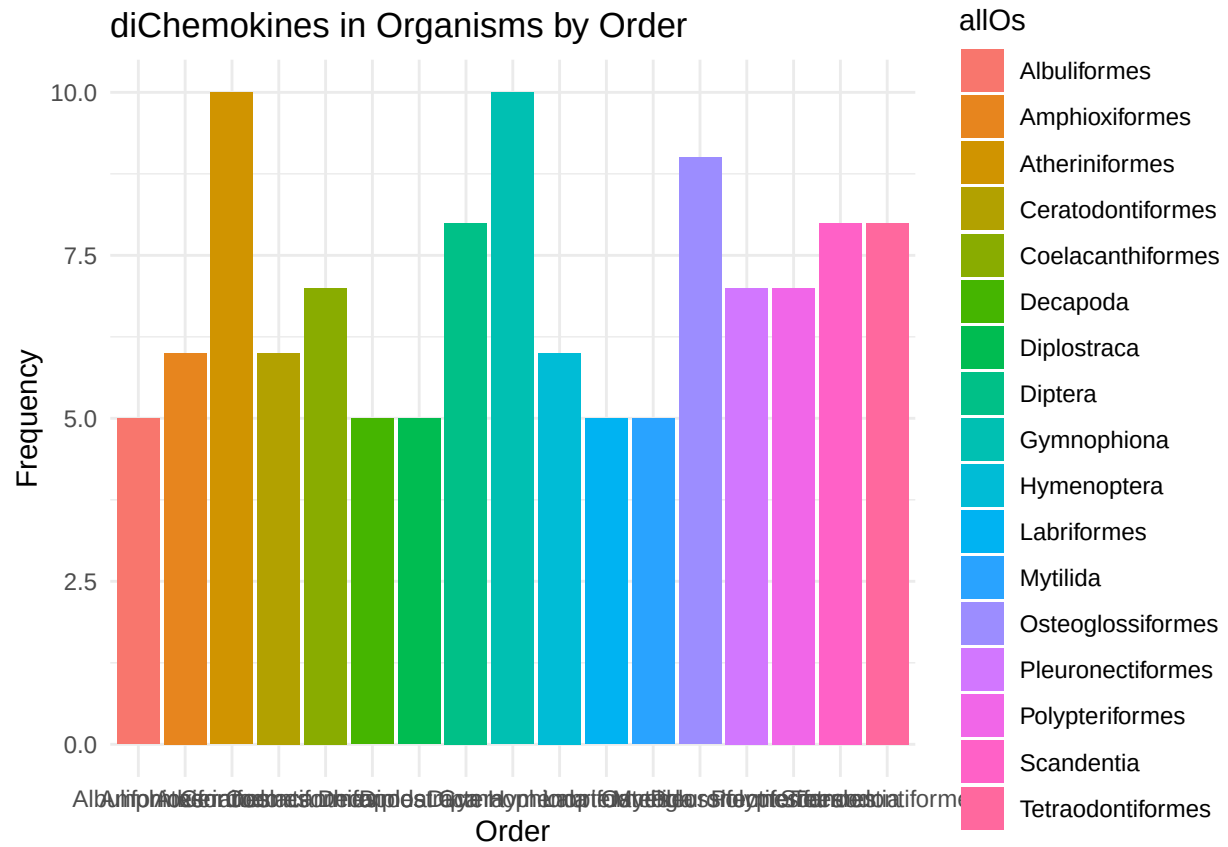


Mid-level taxonomy (Order)

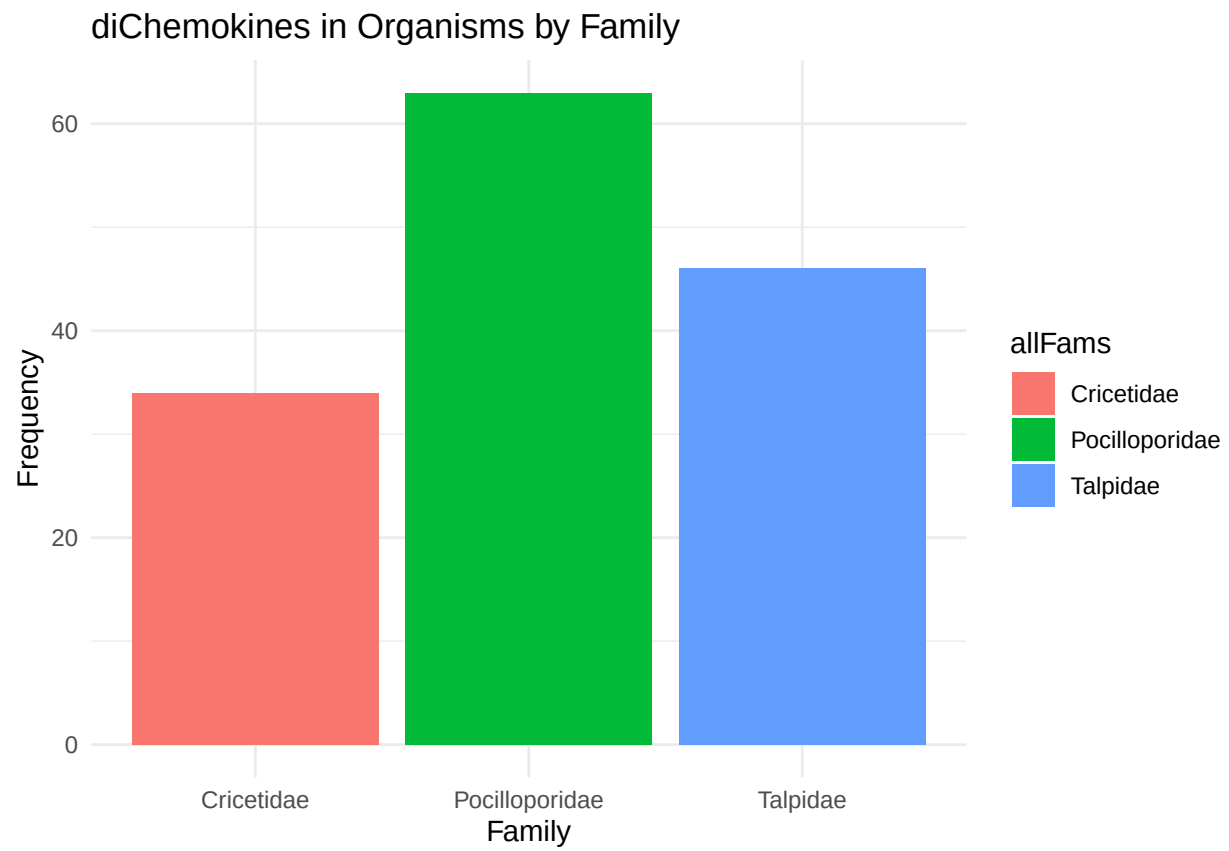
We are considering only order here, and looking for primates in particular.

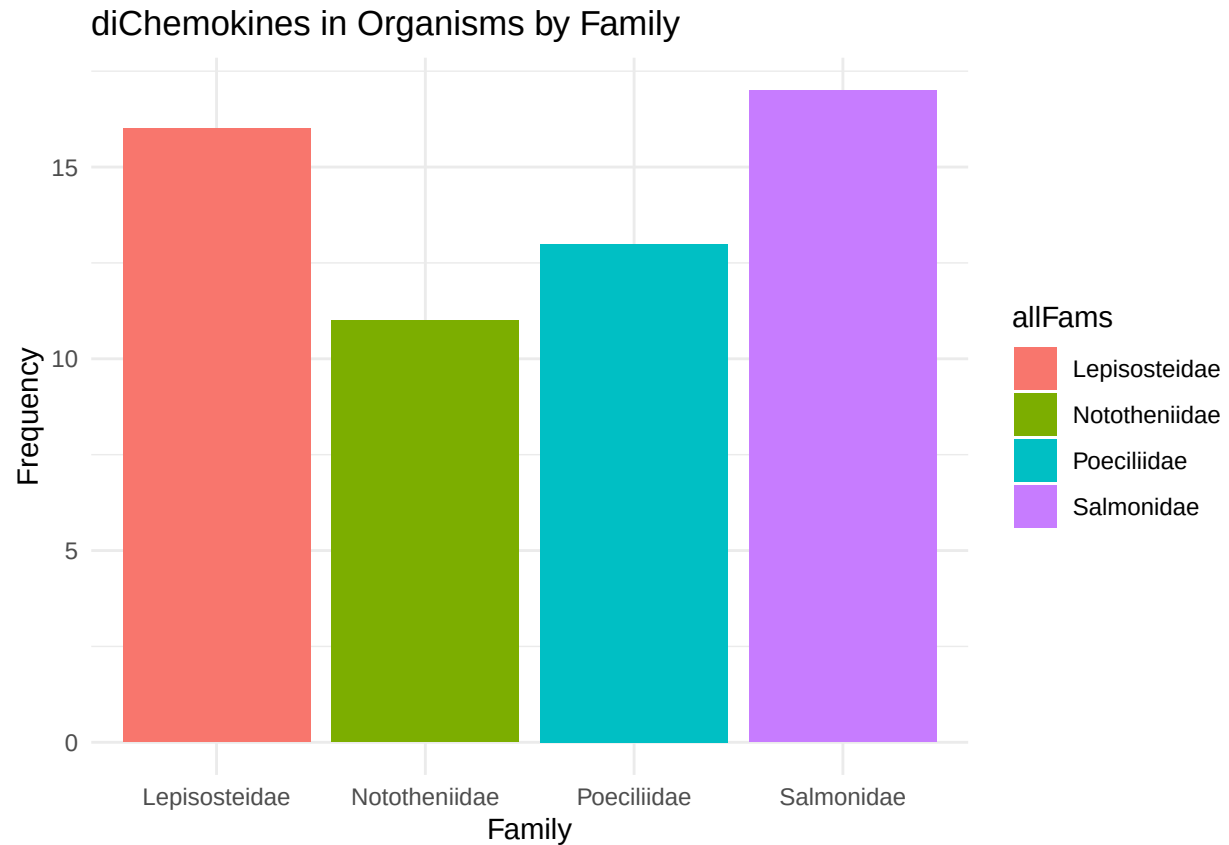






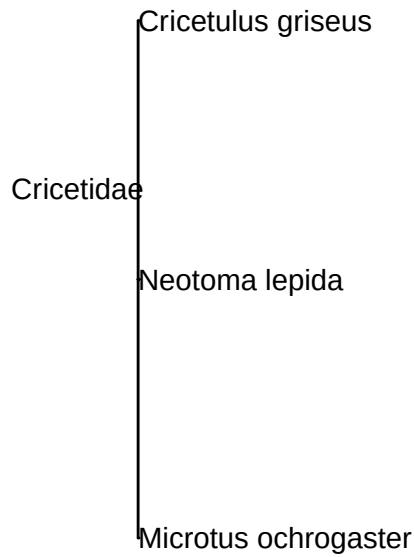
Low-level taxonomy (Family and species)





Cricetidae

```
## Get all ranks and their taxIDs
## Align taxonomy hierarchies...
## Taxonomy alignment done!
## Warning in taxa2dist(df, varstep = varstep, check = check): you used
## 'check=FALSE' and some distances are zero -- was this intended?
## Calculate distance matrix
## Add node labels
```



Families of model organisms

Considered model organisms (and respective families): if they are found

- *Mus musculus* (Rodentia):
- *Rattus norvegicus* (Rodentia):
- *Drosophila melanogaster* (Drosophilidae):
- *Bos taurus* (Artiodactyla):
- *Callosobruchus maculatus* (Coleoptera):
- *Danio rerio* (Cypriniformes):
- *Fundulus heteroclitus* (Cyprinodontiformes) ~not a “model” :
- *Nothobranchius furzeri* (Cyprinodontiformes):
- *Oryzias latipes* (Beloniformes):
- *Anolis carolinensis* (Squamata):
- *Gallus gallus* (Galliformes): (UPI001F01E243 is in *Gallus gallus*?)
- *Taeniopygia castanotis* (Passeriformes):
- *Xenopus laevis* (Anura):
- *Xenopus tropicalis* (Anura):
- *Saccharomyces cerevisiae* (Saccharomycetales):

- Any primate might be good to start with (Primates):
- *Strongylocentrotus purpuratus* (Echinoida):

There are low amount of hits in Muridae, but it can be interesting since it has *Rattus* and *Mus* in it as geni (mice and rats as model organisms)